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Specifications
Building Code Analysis
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MotionAccess, LLC
Attn: Joseph Madden, Managing Partner
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Elk Grove Village, Illinois 60007

Re: Condor Swing: Automatic Door Operator
Review for Installation on Fire Doors
RLGA No. 2012-08

04.06.12

Mr. Madden:

RLGA Technical Services ("RLGA") has reviewed the Condor Swing automatic door operator for installation on fire doors. As a result of this review, RLGA has determined that the **Condor Swing automatic door operator is acceptable for use on fire doors** of any rating required by the *International Building Code* (IBC).

Documents Reviewed

In addition to the codes and standards referenced in this letter, the following product documents were reviewed:

- MET Laboratories, Inc. Certification No. E113016, dated May 24, 2011 (including MET cover letter dated May 23, 2011 and MET Report 28101 dated September 21, 2010).
- Condor Swing, MA50777 Swing Control, Installation and Operation Manual (MA50777-MNL rev P).
- Condor Operator Installation Manual (MA-MANUAL-SWG rev C).
- Push Arm Installation Manual (MA81460-MNL rev 0).
- Pull Arm Installation Manual (MA81470-MNL rev 0).
- Drawings:
 - Simultaneous Pair, Surface Applied, In Swing (Pull Arms) rev D.
 - Simultaneous Pair, Surface Applied, Out Swing (Push Arms) rev D.
 - Single Door, Surface Applied, In Swing (Pull Arm) rev D.
 - Single Door, Surface Applied, Out Swing (Push Arm) rev D.
 - Fire Alarm System Wiring rev A.

Understanding of Door Operator

The Condor Swing is an automatic door operator that can be installed as a low-energy operator per ANSI/BHMA A156.19 or as a high-energy operator per ANSI/BHMA A156.10.

The operator has received certification by MET Laboratories, Inc. for compliance with UL 325.

Overview of Requirements

Definitions (per NFPA 80, *Fire Doors and Other Opening Protectives*):

- **Automatic-Closing Door.** A door that normally is open but that closes when the automatic-closing device is activated.

- **Self-Closing Doors.** Doors that, when opened and released, return to the closed position.
- **Power-Operated Fire Doors.** Doors that normally are opened and closed electrically or pneumatically.
- **Fire Door Hardware.** Door hardware furnished for swinging and sliding fire doors by the door manufacturer as a component part of the listed door assembly.

Since there is no definition of *builders hardware*, it can be defined as any hardware applied to the door that is not considered fire door hardware. An automatic door operator that is not tested and listed as part of a fire door assembly is considered builders hardware.

The IBC does not specifically address power-operated fire doors. However, it does require that fire doors comply with NFPA 80. Additionally, the IBC requires that fire doors be automatic- or self-closing. Power-operated doors, whether connected or disconnected from power, must meet the requirements for automatic- or self-closing doors.

NFPA 80 establishes criteria for builders hardware applied to tested fire doors in the following requirement:

Power-Operated Fire Doors. Power-operated fire doors shall be equipped with a releasing device that shall automatically disconnect the power operator at the time of fire, allowing a self-closing or automatic device to close and latch the door regardless of power failure or manual operation.

Although UL 325 is referenced by the IBC, it is not referenced as a standard for power-operated doors; however, the performance requirements established by the IBC are tested as part of UL 325.

Installation on Fire Doors

For the Condor Swing to be acceptable for use on a fire door, the operator needs to be connected to the fire alarm system. The fire alarm, when activated, must disconnect power from the operator, thus converting the operator into a typical door closer. The fire alarm may be triggered by any type of detection device, manual pull station, or upon the activation of a fire sprinkler system.

Those specifying Condor Swing automatic door operators for fire doors must ensure that other elements of the fire door comply with the requirements of the IBC and NFPA 80, such as:

- Doors shall be side-swinging.
- Doors and frames shall be tested per NFPA 252 or UL 10C.
- Doors and frames shall have a label conforming to the requirements of NFPA 80.
- Doors must have a latch that secures the door when in the closed position.
- Doors must be sealed for smoke and draft control per UL 1784.
- Louvers are prohibited by the IBC.

- Doors must have safety devices as required by ANSI/BHMA A156.10 or ANSI/BHMA A156.19, as applicable.

Inspection and Maintenance

The automatic door operator shall be included in the functional test of fire doors are required by NFPA 80. This should include testing of releasing device to verify disconnection of power.

If you have any questions about the content of this review, please contact me at (602) 569-9645, or at ron@specsandcodes.com.

Sincerely,



Ronald L. Geren, AIA, CSI, CCS, CCCA, SCIP
Owner/Principal

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